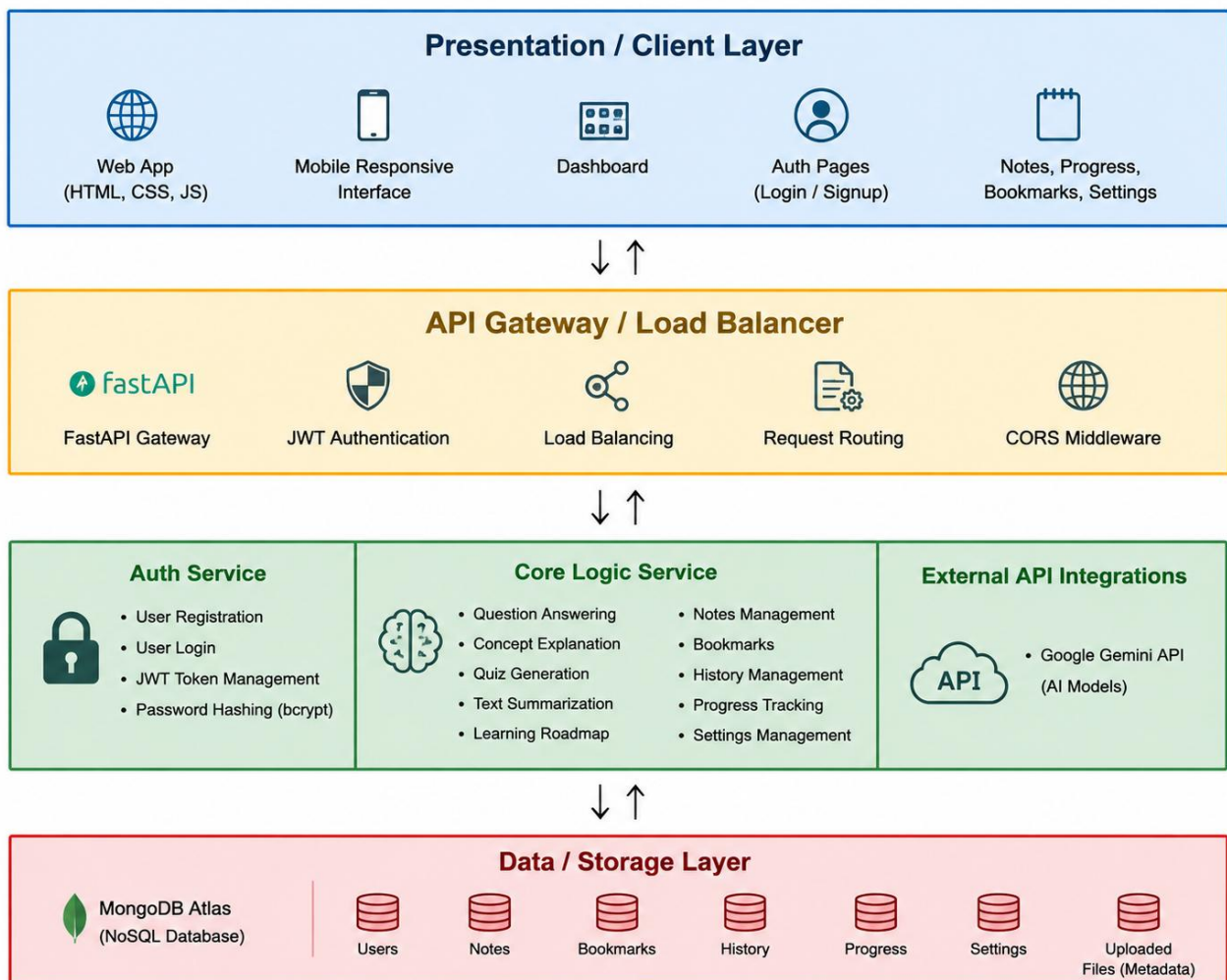


Solution Architecture

Date	26 June 2026
Team ID	-
Project Name	EduGenie – AI-Powered Educational Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for authentication, AI learning tools, notes, progress tracking, bookmarks, and settings.	HTML5, CSS3, JavaScript
API Gateway	Receives client requests, validates them, routes them to appropriate services, and secures endpoints using JWT authentication.	Fast API, REST API, JWT, CORS
Application Services	Implements business logic including AI-powered question answering, concept explanation, quiz generation, summarization, learning roadmap generation, notes, bookmarks, history, and progress tracking.	Python, Fast API, Google Gemini API
Database	Stores user accounts, notes, bookmarks, history, progress, settings, and uploaded file metadata securely.	MongoDB Atlas, Motor, PyMongo